



**NMAM INSTITUTE
OF TECHNOLOGY**

MINI PROJECT REPORT

Setup And Monitoring Honeypot using KFSensor

Subject: Information & Network Security

to

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1. Introduction

In today's rapidly evolving cyber landscape, organizations face an ever-increasing number of sophisticated threats targeting their networks and systems. Traditional security mechanisms such as firewalls, intrusion detection systems (IDS), and antivirus software, while essential, are often reactive in nature — they detect and block known threats but provide limited insight into attacker behaviour and novel attack vectors.

A honeypot is a deliberately deployed decoy system designed to attract and engage malicious actors. Unlike production systems, a honeypot has no legitimate business purpose; any interaction with it is inherently suspicious and potentially malicious. By luring attackers into engaging with a fake environment, security professionals can observe, log, and analyse attack patterns without endangering real assets.

KFSensor is a Windows-based honeypot system developed by KeyFocus Ltd. It simulates a wide range of network services — including HTTP, FTP, SMTP, Telnet, and more — to attract and capture attack traffic. KFSensor provides a rich graphical interface for monitoring live connections, reviewing logged events, and analysing attacker activity in real time. It is widely used in academic and professional security research environments due to its ease of setup and comprehensive logging capabilities.

This experiment involves setting up KFSensor on a Windows machine within a controlled lab environment, configuring it to emulate vulnerable services, and monitoring the resulting network activity to understand attacker behaviour and techniques.

2. Objective

The primary objectives of this experiment are:

- To understand the concept of honeypots and their role in modern network security.
- To install and configure KFSensor as a honeypot on a Windows-based system.
- To simulate vulnerable network services such as HTTP, FTP, Telnet, and SMTP.
- To monitor and log incoming connections and attack attempts in real time.
- To analyze the captured data to identify attack patterns, source IP addresses, and targeted ports.

- To evaluate the effectiveness of honeypots as a proactive cyber-defense and threat intelligence tool.
- To gain hands-on experience with honeypot deployment and network traffic monitoring.

3. Methodology

The experiment was carried out in a structured, step-by-step manner within a controlled lab environment. The following subsections describe the complete methodology adopted.

3.1 Environment Setup

The experiment was conducted using the following hardware and software environment:

Component	Details
Operating System	Windows 10 / Windows 11
Honeypot Software	KFSensor (KeyFocus Ltd.)
Network	Local Area Network (Lab Environment)
Attacker Machine	Kali Linux VM
Tools Used	KFSensor GUI, Command Prompt, Nmap

3.2 Installation of KFSensor

KFSensor was downloaded from the official KeyFocus website and installed on the target Windows machine. During installation, the following steps were performed:

- Downloaded the KFSensor installer (.exe) from the official source.
- Ran the setup wizard and accepted the license agreement.
- Chose the default installation directory and completed the installation.
- Launched KFSensor for the first time to trigger the initial configuration wizard.
- Completed the setup wizard, which automatically configured a default set of simulated services and listeners.

3.3 Configuration of Honeypot Services

After installation, KFSensor was configured to simulate various network services. The following listeners (simulated services) were enabled:

Service	Port	Purpose
FTP	21	Simulate a file transfer server to capture login attempts
Telnet	23	Detect remote access and brute-force login attempts
HTTP	80	Simulate web traffic and data transfer across the World Wide Web.
SMB	445	Used for modern SMB direct-hosted traffic, allowing file sharing over TCP/IP without NetBIOS.

3.4 Network Scanning from Attacker Machine

To simulate a real attack scenario, an attacker machine (Kali Linux / another machine on the same LAN) was used to perform a network scan on the honeypot's IP address using Nmap. The following command was executed from the terminal:

```
nmap -sV 192.168.56.1
```

This command performed service version detection and OS fingerprinting. The scan results revealed the open ports being simulated by KFSensor, allowing us to verify that the honeypot was actively presenting itself as a multi-service system to the network.

3.5 Monitoring & Logging

KFSensor's real-time monitoring dashboard was observed continuously during the experiment. The following monitoring activities were performed:

- Observed the live event log in KFSensor's main GUI window showing incoming connections.
- Reviewed each event entry for details such as source IP, destination port, protocol, and timestamp.
- Examined the packet data captured for individual connection events.
- Used the KFSensor 'Sensor' panel to view which simulated services were receiving traffic.
- Filtered events by severity level: Low, Medium, and High.

4. Results

Upon successful configuration and activation of KFSensor, the honeypot began receiving connection attempts from various sources within the network. The following results were observed and recorded during the experiment.

4.1 Nmap Scan Results

The Nmap scan performed from the attacker machine successfully detected the open ports on the honeypot. The scan output confirmed that KFSensor was presenting itself as a multi-service system with various apparent vulnerabilities. The OS fingerprinting result indicated the honeypot appeared as a Windows machine on the network, consistent with the actual host OS.

```
(kali㉿kali)-[~]
$ ping 192.168.56.1
PING 192.168.56.1 (192.168.56.1) 56(84) bytes of data.
64 bytes from 192.168.56.1: icmp_seq=1 ttl=128 time=2.62 ms
64 bytes from 192.168.56.1: icmp_seq=2 ttl=128 time=1.21 ms
64 bytes from 192.168.56.1: icmp_seq=3 ttl=128 time=1.72 ms
64 bytes from 192.168.56.1: icmp_seq=4 ttl=128 time=1.17 ms
64 bytes from 192.168.56.1: icmp_seq=5 ttl=128 time=1.23 ms
64 bytes from 192.168.56.1: icmp_seq=6 ttl=128 time=1.74 ms
64 bytes from 192.168.56.1: icmp_seq=7 ttl=128 time=1.67 ms
^C
— 192.168.56.1 ping statistics —
7 packets transmitted, 7 received, 0% packet loss, time 6040ms
rtt min/avg/max/mdev = 1.171/1.622/2.615/0.469 ms

(kali㉿kali)-[~]
$ nmap -sV 192.168.56.1
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-23 13:03 -0400
Nmap scan report for 192.168.56.1
Host is up (0.00089s latency).
Not shown: 994 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  tcpwrapped
23/tcp    open  tcpwrapped
80/tcp    open  tcpwrapped
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds?
MAC Address: 0A:00:27:00:00:05 (Unknown)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.90 seconds
```

4.2 Services Successfully Simulated

The honeypot was configured using KFSensor to simulate multiple commonly targeted network services. These services were set up using simulated banners to mimic real server behavior and attract potential attackers.

The following services were successfully simulated:

- **FTP (Port 21):** Configured with a simulated FTP banner, allowing connection attempts and mimicking a real file transfer service.

```
(kali㉿kali)-[~]  
$ ftp 192.168.56.1  
Connected to 192.168.56.1.  
220 192.168.56.1 FTP server (Version wu-2.4.2-academ[BETA-16](1) Thu, 23 Apr 2026 17:05:57 GMT) ready.  
Name (192.168.56.1:kali): admin  
Name (192.168.56.1:(none)):  
ftp: Login failed  
ftp>  
zsh: suspended  ftp 192.168.56.1
```

- **Telnet (Port 23):** Simulated to imitate remote login access, enabling observation of unauthorized access attempts.

```
(kali㉿kali)-[~]  
$ telnet 192.168.56.1  
Trying 192.168.56.1 ...  
Connected to 192.168.56.1.  
Escape character is '^]'.  
Red Hat Linux release 5.1 (Manhattan)  
Kernel 2.0.34 on an i586  
login:Connection closed by foreign host.
```

- **HTTP (Port 80):** Configured to respond to web requests, simulating a basic web server environment.

```
(kali㉿kali)-[~]  
$ curl 192.168.56.1  
  
(kali㉿kali)-[~]  
$ nmap -p 80 --script=http-title 192.168.56.1  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-23 13:08 -0400  
Nmap scan report for 192.168.56.1  
Host is up (0.00067s latency).  
  
PORT      STATE SERVICE  
80/tcp    open  http  
|_http-title: Site doesn't have a title (text/html; charset=utf-8).  
MAC Address: 0A:00:27:00:00:05 (Unknown)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.80 seconds
```

- **SMB (Ports 139 and 445):** Simulated to represent Windows file-sharing services, allowing detection of enumeration attempts.

```
(kali@kali)-[~]
$ smbclient -L 192.168.56.1
Password for [WORKGROUP\kali]:
session setup failed: NT_STATUS_ACCESS_DENIED

(kali@kali)-[~]
$ smbclient -L 192.168.56.1 -N
session setup failed: NT_STATUS_ACCESS_DENIED
```

These services were successfully detected as open during scanning and responded to connection attempts, confirming proper honeypot configuration.

4.3 Captured Events in KFSensor

The KFSensor event log captured multiple connection events during the monitoring period. The results are summarized in the table below:

KFSensor Professional - Evaluation Trial										
File View Scenario Signatures Settings Help										
kfsensor - localhost - honeypot										
	ID	Start	Duration	Protocol	Sensor Port	Name	Visitor	Description	Received	Sig.
21 linuxFTP - Recent Activity	720	23-04-2026 22:41:07.000	1.065	WIN	3389	Logon		auditType:Failure Audit Account:kali...	Audit Type: Failure A...	
23 linuxTelnet - Recent Activity	719	23-04-2026 22:38:54.285	0.252	TCP	80	HTTP Apache	192.168.56.102	GET / HTTP/1.1[OD 0...	[02 01 06 00 C9 0A E...	
80 HTTP Apache - Recent Activity	718	23-04-2026 22:38:54.024	0.253	TCP	80	HTTP Apache	192.168.56.102		[16 03 00 00][01 00 ...	
445 linuxSmbMail - Native service	717	23-04-2026 22:38:53.975	0.001	TCP	80	HTTP Apache	192.168.56.102	Syn Scan With Client Reset. nmap -sS		
	716	23-04-2026 22:38:45.650	0.000	UDP	68	UDP Packet	192.168.31.1		[02 01 06 00 C9 0A E...	
	715	23-04-2026 22:38:13.870	0.000	UDP	68	UDP Packet	192.168.31.1		[02 01 06 00 E5]eM[E...	
	714	23-04-2026 22:37:37.677	0.251	TCP	80	HTTP Apache	192.168.56.102	GET / HTTP/1.1[OD 0...	[FF FD]&[FF FB]&[FF ...	
	713	23-04-2026 22:36:18.398	0.258	TCP	23	linuxTelnet	192.168.56.102		[02 01 06 00 F8]@Aq...	
	712	23-04-2026 22:36:15.258	0.000	UDP	68	UDP Packet	192.168.31.1			
	711	23-04-2026 22:35:57.376	4.003	TCP	21	linuxFTP	192.168.56.102			
	710	23-04-2026 22:35:53.607	0.000	UDP	5775	UDP Packet	192.168.31.51	Broadcast Packet	beacon:RFLSBDLAA0...	
	709	23-04-2026 22:35:51.558	0.000	UDP	5775	UDP Packet	192.168.31.51	Broadcast Packet	beacon:RFLSBDLAA0...	
	708	23-04-2026 22:35:49.511	0.000	UDP	5775	UDP Packet	192.168.31.51	Broadcast Packet	beacon:RFLSBDLAA0...	
	707	23-04-2026 22:35:45.414	0.000	UDP	5775	UDP Packet	192.168.31.51	Broadcast Packet	beacon:RFLSBDLAA0...	
	706	23-04-2026 22:35:43.366	0.000	UDP	5775	UDP Packet	192.168.31.51	Broadcast Packet	beacon:RFLSBDLAA0...	
	705	23-04-2026 22:35:41.318	0.000	UDP	5775	UDP Packet	192.168.31.51	Broadcast Packet	beacon:RFLSBDLAA0...	
	704	23-04-2026 22:35:37.216	0.000	UDP	5775	UDP Packet	192.168.31.51	Broadcast Packet	beacon:RFLSBDLAA0...	
	703	23-04-2026 22:35:35.175	0.000	UDP	5775	UDP Packet	192.168.31.51	Broadcast Packet	beacon:RFLSBDLAA0...	
	702	23-04-2026 22:35:33.228	0.000	UDP	5775	UDP Packet	192.168.31.51	Broadcast Packet	beacon:RFLSBDLAA0...	
	701	23-04-2026 22:34:54.337	0.000	UDP	68	UDP Packet	192.168.31.1		[02 01 06 00 D6 9B E...	
	700	23-04-2026 22:34:54.332	0.000	UDP	68	UDP Packet	192.168.31.1		[02 01 06 00 D6 9B E...	
	699	23-04-2026 22:34:15.881	0.000	UDP	5775	UDP Packet	192.168.31.51	Broadcast Packet	beacon:RFLSBDLAA0...	
	698	23-04-2026 22:34:13.834	0.000	UDP	5775	UDP Packet	192.168.31.51	Broadcast Packet	beacon:RFLSBDLAA0...	
	697	23-04-2026 22:34:11.785	0.000	UDP	5775	UDP Packet	192.168.31.51	Broadcast Packet	beacon:RFLSBDLAA0...	
	696	23-04-2026 22:34:09.736	0.000	UDP	5775	UDP Packet	192.168.31.51	Broadcast Packet	beacon:RFLSBDLAA0...	
	695	23-04-2026 22:33:28.336	0.000	TCP	8083	TCP Closed Port	192.168.56.102			
	694	23-04-2026 22:33:28.336	0.000	TCP	30718	TCP Closed Port	192.168.56.102			
	693	23-04-2026 22:33:28.336	0.000	TCP	524	TCP Closed Port	192.168.56.102			
	692	23-04-2026 22:33:28.336	0.000	TCP	3493	TCP Closed Port	192.168.56.102			
	691	23-04-2026 22:33:28.336	0.000	TCP	32775	TCP Closed Port	192.168.56.102			
	690	23-04-2026 22:33:28.336	0.000	TCP	340	TCP Closed Port	192.168.56.102			
	689	23-04-2026 22:33:28.336	0.000	TCP	1023	TCP Closed Port	192.168.56.102			
	688	23-04-2026 22:33:28.336	0.000	TCP	3527	TCP Closed Port	192.168.56.102			
User Rights: Admin [78] Server: Running Visitors: 11 Events: 709/709										

5. Observations

5.1 Honeypot Effectiveness

The honeypot was highly effective in attracting and logging simulated attack traffic within the lab environment. KFSensor accurately captured all connection attempts made to its simulated services and provided detailed logs with timestamps, source IP addresses, and packet data.

5.2 Attack Behavior Patterns

During the experiment, various attack techniques were performed from Kali Linux, including port scanning and protocol-specific connection attempts. The honeypot logs generated by KFSensor revealed the following behavior patterns:

- **Port Scanning Activity:**
Nmap scans generated multiple connection attempts across different ports in a short time, indicating reconnaissance activity typically performed by attackers to identify open services.
- **Multi-Port Probing:**
The attacker system attempted connections to several common service ports (FTP, Telnet, HTTP, SMB), suggesting an effort to discover vulnerable entry points.
- **Repeated Connection Attempts:**
Certain services, especially FTP and Telnet, showed repeated access attempts, indicating possible brute-force or service probing behavior.
- **Protocol-Specific Interactions:**
Different attack tools triggered unique responses:
 1. FTP attempts initiated banner exchange
 2. Telnet attempts resulted in session initiation
 3. HTTP requests generated web access logs
 4. SMB attempts triggered authentication prompts
- **Source Identification:**
All attack attempts were consistently traced back to the attacker machine's IP address, demonstrating the honeypot's ability to log and identify intrusion sources.
- **Effect of Configuration:**
It was observed that proper configuration (active ports, simulated banners, and disabled firewall) was essential for capturing complete attack data.

5.3 Severity Distribution

The events captured by KFSensor were classified into severity levels as follows:

- **High Severity:** Connection attempts on FTP and Telnet — services commonly exploited for unauthorized access.
- **Medium Severity:** HTTP, SMB connection attempts, which could indicate web crawlers, vulnerability scanners, or spam relay attempts.
- **Low Severity:** General port probes and SYN packets that did not proceed beyond the initial connection handshake.

5.4 Key Learnings

This experiment provided several important insights into honeypot deployment and network security monitoring:

- Honeypots are effective tools for passively gathering threat intelligence without exposing real production systems.
- Even within a controlled lab environment, simulated services attract a wide variety of connection attempts when visible on the network.
- KFSensor's detailed logging capabilities make it easy to correlate events, identify attacker IP addresses, and study attack sequences.
- The use of decoy services can significantly slow down an attacker by engaging them with non-productive systems.
- Regular analysis of honeypot logs can help organizations identify new and emerging attack techniques before they target real assets.

6. Conclusion

This experiment successfully demonstrated the deployment and monitoring of a honeypot using KFSensor in a controlled lab environment. The honeypot effectively simulated multiple vulnerable network services and captured a variety of attack attempts including port scans, brute-force login attempts, SMB enumeration, and web vulnerability probes.

KFSensor proved to be a powerful and user-friendly honeypot solution, offering real-time monitoring, detailed event logging, and severity classification. The experiment reinforced the importance of honeypots as a complementary security mechanism alongside traditional defensive tools. By analyzing attacker behavior in a safe environment, security teams can proactively improve their defenses and stay ahead of evolving threats.

In conclusion, honeypots like KFSensor play a critical role in threat intelligence gathering, early intrusion detection, and cybersecurity research. The knowledge gained from this experiment is directly applicable to real-world network security scenarios and provides a strong foundation for further study in the field of information and network security.

7. References

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